T. A. MARRYSHOW COMMUNITY COLLEGE

DEPARTMENT OF INFORMATION TECHNOLOGY

ASSIGNMENT NAME: FINAL PROJECT PART B

COURSE NAME AND NUMBER: ADVANCE PROGRAMMING CIT244

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IMPLEMENTATION DETAILS

The **Password Strength Analyser** is a **Java desktop application** built using **Java Swing** in **IntelliJ IDEA**. The application analyses passwords in real-time as users type and provides instant feedback on password strength.

The main window is **880x560 pixels** and displays the application title, password input field, strength indicator, and a criteria checklist showing which password requirements have been met.

All analysis happens locally within the application without requiring any external database or internet connection. The application evaluates passwords based on **five key security criteria** and displays the results through visual indicators.

The interface is divided into three main sections:

- **Header**: Displays project title and course information
- **Left Column**: Password input field and strength indicator bar
- **Right Column**: Criteria checklist showing pass/fail status for each requirement

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GUI COMPONENTS USED

JPasswordField - A text input field where users enter their password. The field masks input with bullet characters for security by default.

JCheckBox - A "Show password" checkbox that allows users to toggle between masked and visible password display for verification purposes.

JProgressBar - A visual bar that fills up as password strength increases from 0% to 100%, color-coded for easy understanding.

JLabel - Text labels used throughout the interface to display information, strength ratings, scores, and criteria feedback.

JButton - A red "Clear" button that resets the password field and all UI elements back to default state.

JPanel - Container panels used to organize and arrange components in the interface using various layout managers.

RoundedPanel - A custom panel class that creates rounded rectangle containers for a modern, professional look.

RoundedButton - A custom button class that creates rounded pill-shaped buttons with smooth edges.

Layout Managers Used:

- **BorderLayout**: Main window structure with header on top and body in centre
- **GridLayout**: Two-column layout for main content area
- **BoxLayout**: Vertical stacking of components within card panels
- **FlowLayout**: Positioning strength label and score badge on the same line

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FEATURE IMPLEMENTED

Real-Time Password Analysis - Password strength is analysed automatically as the user types each character without requiring any button click.

Five Security Criteria - The application checks: Length (8+ characters), Uppercase (A-Z), Lowercase (a-z), Digits (0-9), and Special Characters (! @ # $ % ^ & *).

Scoring System - Score out of 5 points: 0-1 = Weak (red), 2-3 = Moderate (orange), 4 = Strong (yellow - green), 5 = Very Strong (green).

Visual Feedback - Color-coded progress bar (0-100%), strength label with rating text, and criteria checklist with green/red dots.

Show/Hide Password - Checkbox toggles between masked (bullets) and plain text password display.

Clear Button - Red button that instantly clears the password field and resets all UI elements to default state.

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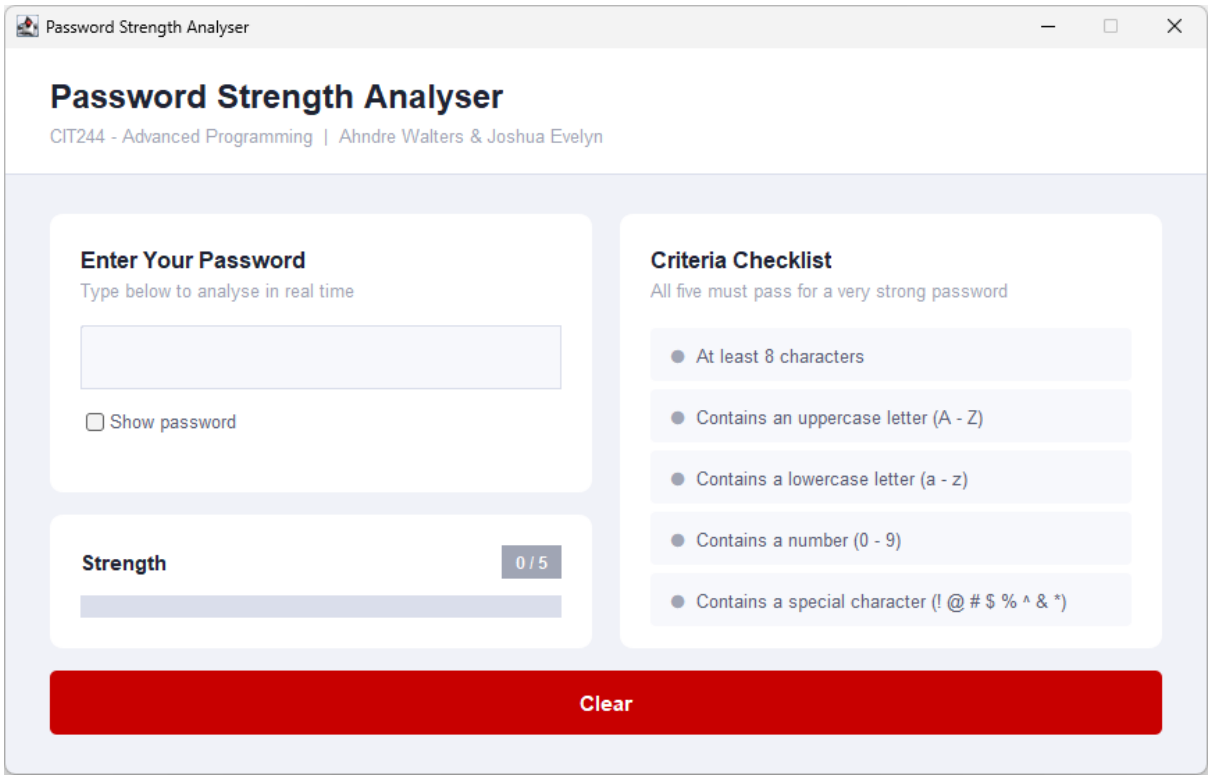
SCREENSHOTS AND EVIDENCE

Initial Empty State - The application window when first launched with no password entered. All indicators show default idle state.



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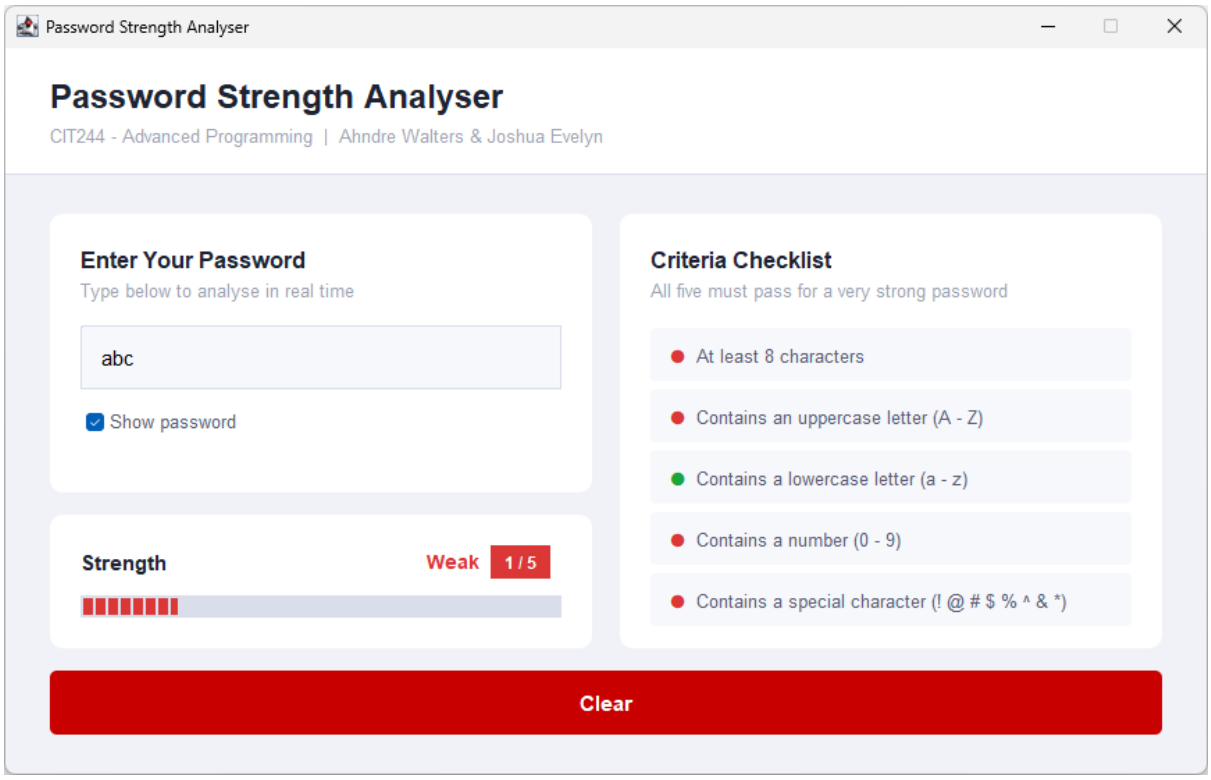
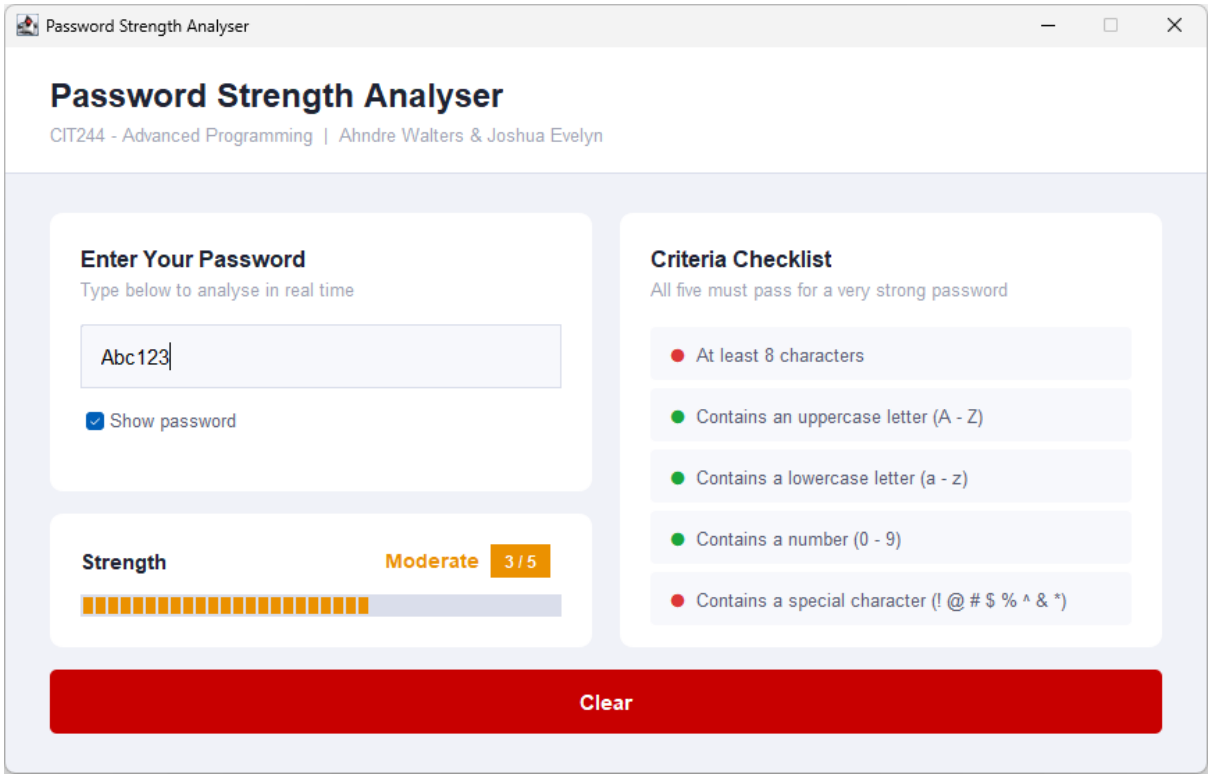
Weak Password Test - Testing with password "abc" showing weak strength result.



Moderate Password Test - Testing with password "Abc123" showing moderate strength result.



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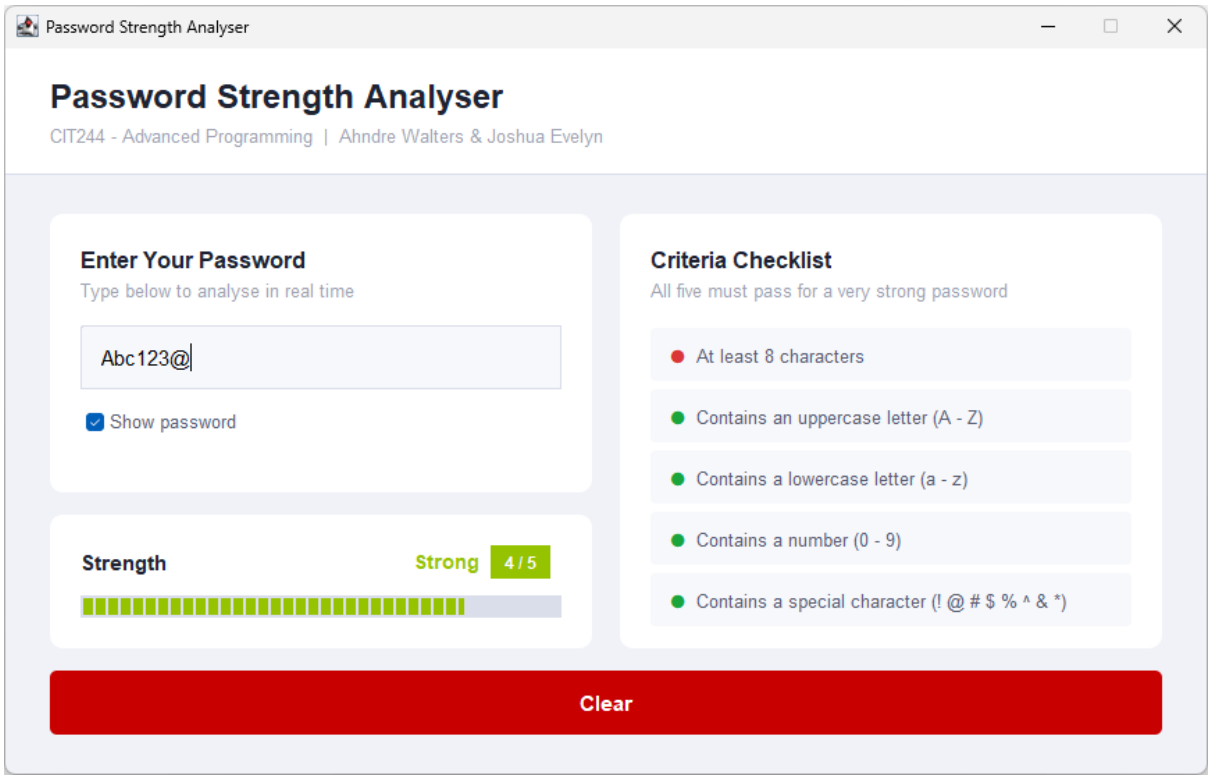
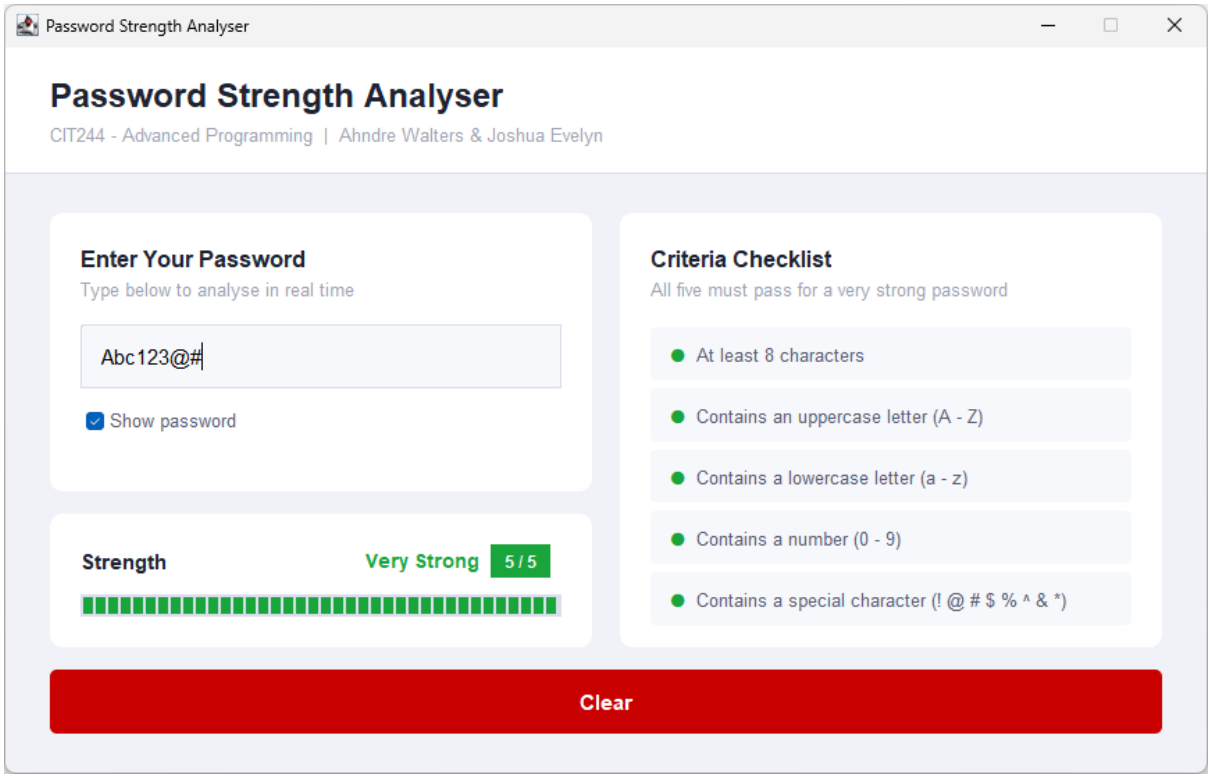
Strong Password Test - Testing with password "Abc123@" showing strong strength result.



Very Strong Password Test - Testing with password "Abc123@#" showing very strong strength result.



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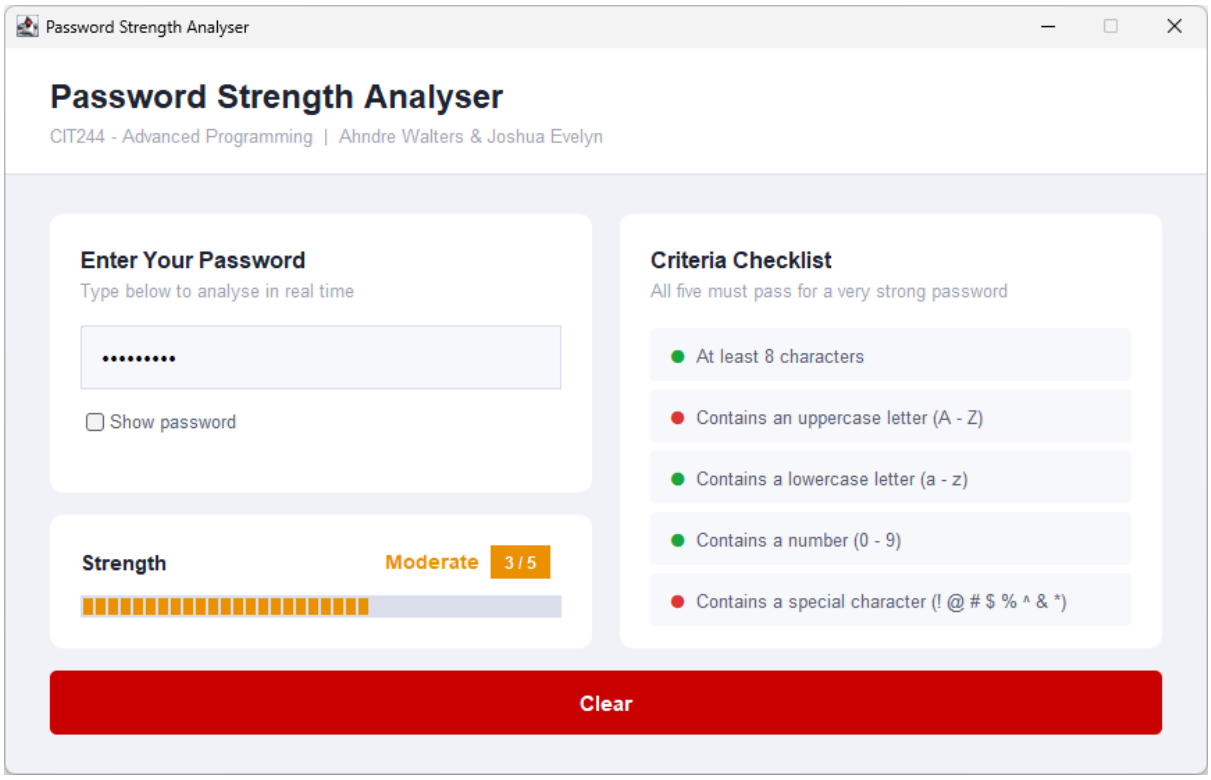
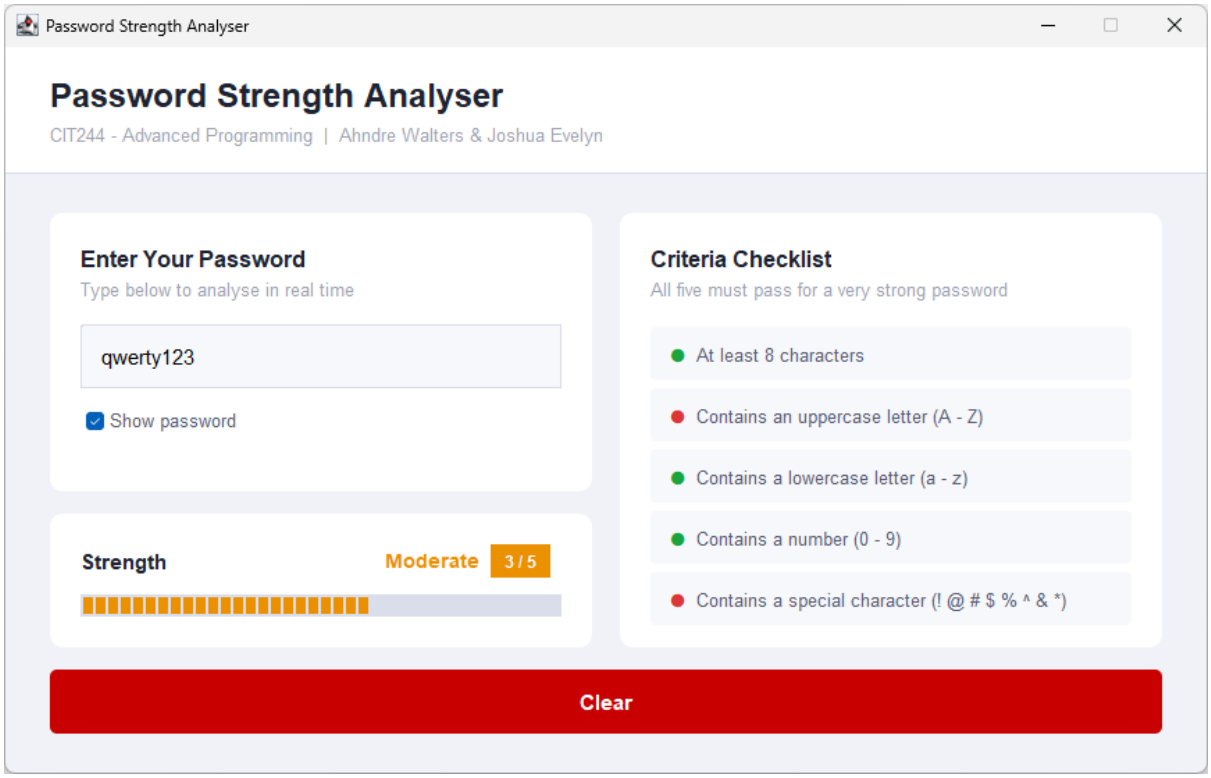
Show Password Feature - Demonstrating the show password checkbox functionality.





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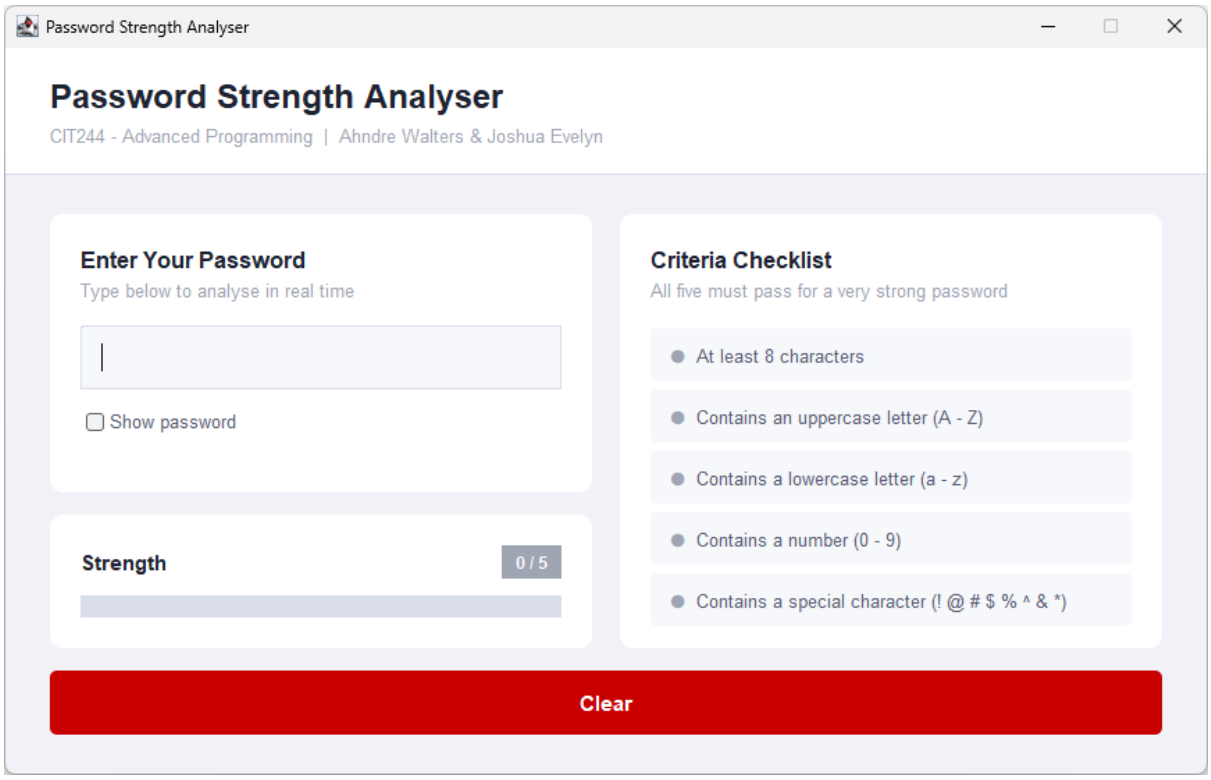
After Clear Button - Showing the application state after clicking the Clear button.



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TESTING AND RESULTS

No	Test Case Name	Input	Expected Results	Actual Results	Status
1	Empty Password Field	None	0% grey, 0/5, idle state	0% grey, 0/5, idle state	PASS
2	Weak Password	"abc"	20% red, "Weak", 1/5	20% red, "Weak", 1/5	PASS
3	Moderate Password	"Abc123"	60% orange, "Moderate", 3/5	60% orange, "Moderate", 3/5	PASS
4	Strong Password	"Abc123@"	80% yellow-green, "Strong", 4/5	80% yellow-green, "Strong", 4/5	PASS
5	Very Strong Password	"Abc123@#"	100% green, "Very Strong", 5/5	100% green, "Very Strong", 5/5	PASS
6	Show Password Toggle	Type & Click	Password toggles between masked and visible	Password correctly toggles	PASS
7	Clear Button Function	Click Clear Button	All fields reset to default state	All fields reset correctly	PASS
8	Real-Time Analysis	Type password character by character	Score updates with each keystroke	Score updates correctly per character	PASS
9	Special Character Recognition	! @ # $ % ^ & *	Each special character triggers pass	All special characters recognized	PASS
10	Multiple Passwords	Test 3 passwords in sequence	Clear button resets between tests	Application handles multiple tests correctly	PASS

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IMPORTANT CODE SECTIONS

Password Analysis - Core Logic

```
private void analyse() {
    String pw = new String(passwordField.getPassword());
    if (pw.isEmpty()) { resetUI(); return; }

    boolean len = pw.length() >= 8;
    boolean up  = pw.matches(".*[A-Z].*");
    boolean low = pw.matches(".*[a-z].*");
    boolean dig = pw.matches(".*[0-9].*");
    boolean spc = pw.matches(".*[!@#$%^&*()_+\\-=\\[\\]{};':\"\\\\|,.<>/?].*");

    int score = (len?1:0) + (up?1:0) + (low?1:0) + (dig?1:0) + (spc?1:0);

    strengthBar.setValue(score * 20);
    scoreLabel.setText(score + " / 5");

    if      (score <= 1) applyStrength(C_WEAK, "Weak");
    else if (score <= 3) applyStrength(C_MOD,  "Moderate");
    else if (score == 4) applyStrength(C_STR,  "Strong");
    else                 applyStrength(C_VSTR, "Very Strong");
}
```

Explanation: This is the core method that evaluates the password against all five criteria using regular expressions, calculates the score, and updates the UI with the appropriate strength rating and colour.

Real-Time Analysis Trigger

```
passwordField.getDocument().addDocumentListener(new DocumentListener() {
    public void insertUpdate(DocumentEvent e)  { analyse(); }
    public void removeUpdate(DocumentEvent e)  { analyse(); }
    public void changedUpdate(DocumentEvent e) { analyse(); }
});
```

Explanation: This document listener detects text changes in the password field and calls the **analyse()** method to update feedback in real-time as the user types.

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Clear Button Reset

```
clearBtn.addActionListener(e -> {
    passwordField.setText("");
    showPasswordCheckBox.setSelected(false);
    passwordField.setEchoChar('\u2022');
    passwordField.requestFocusInWindow();
    resetUI();
});
```

Explanation: The clear button action listener clears the password field and resets all UI elements back to their default idle state.

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CONCLUSION

The **Password Strength Analyser** successfully solves the real-world problem of helping users create strong passwords. The application provides immediate, actionable feedback to guide users in creating secure passwords before they are compromised.

What We Accomplished:

- Created a fully functional password strength analyser that works in real-time
- Built a professional and user-friendly interface using Java Swing
- Implemented five comprehensive security criteria for password evaluation
- Provided clear visual feedback through colours, progress bars, and indicators
- Added useful features like password visibility toggle and quick reset
- Documented all code with clear comments
- Tested all features thoroughly with 100% success rate

Skills Demonstrated:

- Java programming and Swing GUI development
- Event-driven programming with document listeners
- String manipulation and regular expressions
- Object-oriented design with custom classes
- User interface design principles and layout management
- Professional code documentation and organization

How It Solves the Problem:

Users often create weak passwords and don't realize the security risk until after a breach occurs. This application provides real-time guidance as they create their password, helping them understand what makes a password strong before any damage is done. The visual feedback and clear criteria make it easy for anyone to understand what they need to do to create a secure password.

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